

CPSC 304 Project Cover Page

Milestone #: 1

Date: Feb. 2nd, 2026

Group Number: 26

Name	Student Number	CS Alias (Userid)	Preferred E-mail Address
Mark Skrypnyk	86867629	p5k9k	skrypnykmark@gmail.com
Chaitanya Thakral	22283816	r1l4x	cthakral6@gmail.com
Adi Yadav	80981251	b4l2k	aditya_1007@icloud.com

By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia.

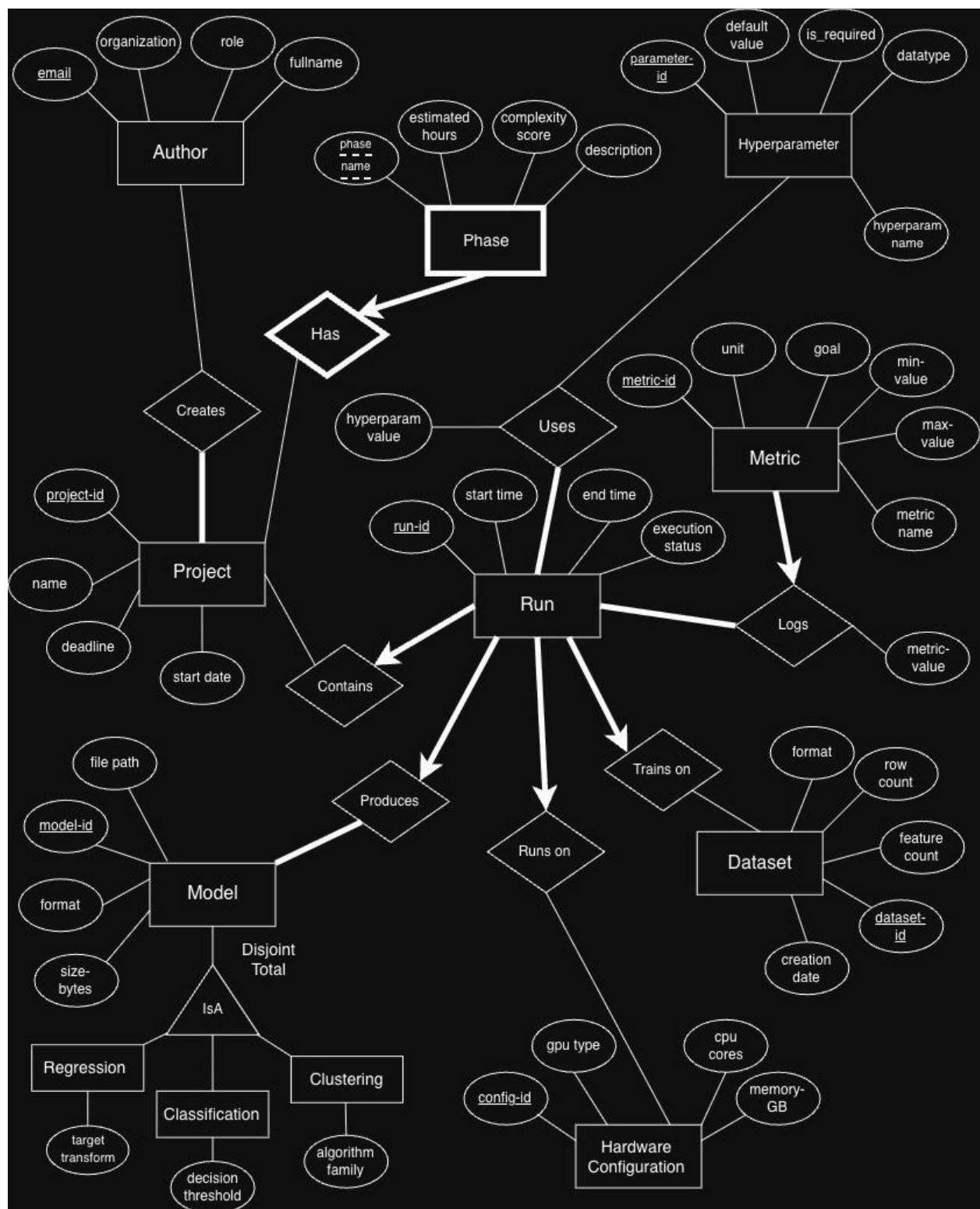
What is the domain of the application? Describe it.

The domain of this application is machine learning experiment tracking and model management. The goal is to support data scientists by storing structured information about how their models are trained and evaluated over many experimental runs. In real machine learning work, data scientists run lots of training experiments where they change things like the dataset, hyperparameters, and computing setup to see what improves performance. The problem is that without a structured system, it's easy to lose track of which run produced which results and how a model was created. This application focuses on organizing those experiments so runs are traceable, comparable, and easier to reproduce.

What aspects of the domain are modeled by the database (functionality)?

The database models the core pieces of a machine learning experimentation workflow. A user (author) creates projects and each project can be broken down into phases like data preparation, training, evaluation with planning details such as estimated hours and complexity. The main tracked unit is a run, which records execution details like the start/end time and status for a single experiment attempt. Each run trains on a dataset, runs on a specific hardware configuration, uses hyperparameters that are defined before the run, and logs metrics to measure performance. Runs also produce models, and the model can be categorized into common types like classification, regression, or clustering. In a real setting, this database could be used by a research lab or machine learning team to track progress, compare experiments, and reproduce results.

ER DIAGRAM NEXT PAGE



Entity set	Meaning	Primary key	Non Key Attributes
Author	A person who creates projects.	Email	- Organization - Role - Full Name
Project	A place to keep and organize all experimental work.	Project ID	- Name - Deadline - Start Date
Run	One execution of the training code.	Run ID	- Start Time - End Time - Execution Status: Pass/Fail represents whether run was successful or not
Dataset	A collection of data points organized for model training.	Dataset ID	- Format: represents file format (CSV , JSON etc.) - Row Count - Feature Count - Creation Date
Metric	An evaluation value tracked for runs.	Metric ID	- Metric name - Unit: what the metric is measured in (percentage, seconds, etc.) - Goal: the optimization target - Min/Max value: expected valid range for the metric which is useful for detecting impossible results.
Hardware Configuration	Hardware details of the computer used.	Config ID	- GPU type: Name of GPU model (NVIDIA A100, NVIDIA RTX3060) - CPU cores: Number of CPU cores available. - Memory GB: Amount of RAM in GB.
Hyperparameter	A configuration setting defined before a run.	Parameter ID	- Hyperparameter name - Datatype: int, float, boolean etc. - Default value: value used for the run if not explicitly defined - Is Required: if the run must define a value
Model	The trained model produced after the training run.	Model ID	- File Path: location on the cloud where the model is saved. - Format: format of the saved model (Pickle, Joblib) - Size Bytes: the size of the model in bytes.
Regression	Model that predicts	—	Target Transform: processing

	continuous numerical value		step where a function is applied to the target variable (like log, transform).
Classification	Model that predicts categorical label.	—	Decision Threshold: numerical cutoff used to interpret the classification model.
Clustering	Model that groups similar data points into clusters.	—	Algorithm Family: represents the kind of clustering model used (KMean , GMM etc.)
Phase	Stage of the project lifecycle.	Phase Name + Project ID	- Estimated hours: predicted time to complete that phase. - Complexity Score: numerical score indicating the difficulty of the phase. - Description: represents the goal of the phase.

AI TOOLS WERE NOT USED IN THIS ASSIGNMENT.